



Structural Mapping for NTSB Incident Retrieval

Improving retrieval through causal chain alignment

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The Problem

1

Embeddings match on surface words - two incidents with "engine, oil, failure" score 0.94 similarity even when the actual failure mechanisms are completely different

2

A maintenance error causing oil starvation and a fatigue crack causing an engine fire look nearly identical to embeddings, but are structurally very different incidents

3

We need a way to compare the causal structure of incidents, not just the words used to describe them

Our Approach: A0 vs A2

A0

Embedding Only

- 1 Cosine similarity on text-embedding-3-small vectors
- 2 Captures word-level and topic-level similarity
- 3 This is our baseline - what the system does today

A2

Structural Mapping

- 1 Embeddings + causal chain alignment scoring
- 2 Matches on structural roles, not surface keywords
- 3 This is the new contribution - grounded in Gentner's theory

Same train/test split for both. Only the retrieval scoring changes.

Step 1: Extract Causal Chains

NTSB Incident Report

Narrative: *"Sudden loss of oil pressure during cruise. Engine seized 2 minutes later..."*

Probable Cause: *"Mechanic failed to torque oil drain plug during annual inspection"*



Extracted Causal Chain

- | | | |
|---|-------------------|---------------------------|
| 1 | initiating_event | maint / maint_error |
| 2 | propagation | lubrication / deprivation |
| 3 | system_compromise | engine / overheating |
| 4 | terminal_failure | engine / separation |
| 5 | outcome | aircraft / power_loss |

- 1 Each step has a role (what function it serves), system (what component), and mechanism (how it causes the next step)
- 2 LLM extracts from controlled vocabularies - 7 roles, 14 systems, 17 mechanisms - so every incident uses the same terms
- 3 This gives us a structured representation we can align mathematically, not just compare words

Formula 1: Element Similarity

$$\text{sim}(a, b) = 0.50 \times \text{role} + 0.30 \times \text{mechanism} + 0.20 \times \text{system}$$

What each part means:

role (50%)

What function does this step serve? (initiating event, propagation, failure, outcome). Weighted highest because the causal function is the deepest structural feature.

mechanism (30%)

How does this step cause the next? (fatigue, overheating, contamination). This is the causal connection between steps.

system (20%)

What physical component? (engine, lubrication, fuel). Weighted lowest - two incidents can involve different parts but share the same failure pattern.

Where this comes from: Gentner (1983) established that deep similarity is about relational structure, not surface features. The 50/30/20 split follows her hierarchy: systematic relations > first-order relations > attributes. Spencer-Smith & Goldstone (1997) showed features are dynamically weighted during comparison.

Formula 1 in Action

Element A (Incident A, Step 1)

role: initiating_event **system:**
maintenance
mechanism:
maintenance_error

Element B (Incident B, Step 1)

role: initiating_event **system:**
engine_mechanical
mechanism:
fatigue

Computing the similarity:

	Role:	Both are "initiating_event" = exact match = 0.50
	Mechanism:	"maintenance_error" vs "fatigue" = not related = 0.00
	System:	"maintenance" vs "engine_mechanical" = not related = 0.00
	Total:	$\text{sim} = 0.50 + 0.00 + 0.00 = 0.50$ (partial match – same role, different details)

Takeaway: Same causal role (both are initiating events) scores high, even though the specifics differ. This is what structure-mapping means -matching on function, not surface.

Formula 2: Chain Alignment

```
dp[i][j] = max(  
    dp[i-1][j-1] + sim(A[i], B[j]),    // align these two  
    dp[i-1][j]   + gap_penalty,        // skip step in A  
    dp[i][j-1]   + gap_penalty        // skip step in B)
```

What each part means:

dp[i][j] Best alignment score for the first i steps of chain A and first j steps of chain B

sim(A[i], B[j]) Element similarity from Formula 1 — how well do these two steps match?

gap_penalty = -0.05 Small cost for skipping a step (one incident has a step the other doesn't)

Where this comes from: Needleman & Wunsch (1970) developed this dynamic programming algorithm for aligning protein sequences in bioinformatics. We use it to align causal chains while enforcing Gentner's constraints: one-to-one mapping (each step maps to at most one), and parallel connectivity (alignment preserves causal order).

Formula 2 in Action

Incident A Step	Incident B Step	Sim	Match Type	Why
initiating_event	initiating_event	0.50	Partial	Same role, diff mech
propagation	propagation	0.50	Partial	Same role, diff mech
system_compromise	system_compromise	0.50	Partial	Same role, diff detail
(skipped)	propagation (#4)	GAP	Unmapped	Extra step in B
terminal_failure	terminal_failure	0.70	Strong	Role + mech match
outcome	outcome	0.50	Partial	Same role

- 1 The algorithm lined up 5 pairs and found 1 gap (Incident B has an extra propagation step)
- 2 Most matches are partial (same role but different mechanisms) -structurally similar positions, different specifics
- 3 One strong match (terminal_failure) where both role AND mechanism aligned

Formula 3: Structural Score

$$\text{score} = (\text{strong} + 0.5 \times \text{partial}) / \text{total} - 0.10 \times (\text{unmapped} / \text{total})$$

What each part means:

strong	Pairs with $\text{sim} \geq 0.70$ (role AND mechanism matched). Gets full credit.
partial	Pairs with $0.35 \leq \text{sim} < 0.70$ (role matched but details differ). Gets half credit.
unmapped	Steps with no counterpart in the other chain. Penalized at 0.10 per fraction.
total	Total unique steps across both chains — normalizes the score to $[0, 1]$.

Where this comes from: This operationalizes the Structural Evaluation Score from the Structure-Mapping Engine (Falkenhainer, Forbus & Gentner, 1989). In SME, match hypotheses get evidence values (~ 0.4 - 0.5), and higher-order structure gets trickle-down bonuses. Our strong/partial/unmapped maps directly to their evidence framework.

Formula 3 in Action

From our alignment example:

Count from alignment: `strong = 1, partial = 4, unmapped = 1, total = 10`

Apply the formula: `(1 + 0.5 x 4) / 10 - 0.10 x (1 / 10) = 0.30 - 0.01 = 0.29`

Final structural score: `0.29 (out of 1.0 – these incidents share causal structure but are not close matches)`

- 1 A score of 0.29 means modest structural similarity - the chains share the same general progression but through different mechanisms
- 2 Compare: two incidents with identical causal chains would score ~0.90+, and completely different incidents would score near 0
- 3 The score penalizes the gap (extra step in B) but not harshly - gaps happen in real incidents and shouldn't dominate

Formula 4: Combining with Embeddings

$$w' = \max(0, \text{cosine}) \times \exp(a \times \text{structural_score})$$

What each part means:

w'	The final reranked score used to order retrieval results
cosine	Embedding cosine similarity - the fast first-pass score (how similar the text is)
max(0, ...)	Ensures we never promote something that isn't semantically related at all
exp(a x s)	Exponential amplifier - high structural scores get a big boost, low scores barely change
a (alpha)	Controls reranking strength. a=0: pure embedding. a=2: structural has real impact (our default)

Where this comes from: Two-stage retrieval fusion is standard in information retrieval (Nogueira & Cho, 2019). Fast retriever (embeddings) finds candidates, expensive reranker (structural mapping) reorders them. Goldstone (1994) supports this - similarity and mapping are computed simultaneously, not independently.

Formula 4 in Action

Our example incidents:

Embedding similarity: $\text{cosine} = 0.94$ (embeddings think these are very similar)

Structural score: $s = 0.29$ (causal chains are only modestly similar)

Fusion ($\alpha = 2.0$): $w' = \max(0, 0.94) \times \exp(2.0 \times 0.29) = 0.94 \times 1.79 = 1.68$

Now compare with a truly similar incident (same mechanism):

$\text{cosine} = 0.82, \quad s = 0.85, \quad w' = 0.82 \times \exp(2.0 \times 0.85) = 0.82 \times 5.47 = 4.49$

Result: The structurally similar incident (score 4.49) jumps above the surface-similar one (1.68), even though it had a lower embedding score. The structural mapping corrected the ranking - exactly what we want.

How We Evaluate

Metric	What It Measures
Top-1 Accuracy	Is the best retrieved incident the correct match?
Recall@5	Is the correct match somewhere in the top 5 results?
MRR	Mean Reciprocal Rank - how far down before you find the right answer?
F1 Score	Balance of precision and recall across all retrieval results

- 1 Same train/test split for A0 and A2 - only the scoring method changes between conditions
- 2 All weights and alpha tuned only on training data. Test set held out until final evaluation. No data leakage.
- 3 If A2 doesn't beat A0, that's still a finding - it means embeddings already capture enough structural info for this domain

Status & Next Steps

- ✓ Theoretical framework defined (6 papers, 4 formulas, all from published literature)
 - ✓ Controlled vocabularies built from NTSB classification system
 - ✓ Schema extraction + validation pipeline working (with retry logic)
 - ✓ struct_score_v2.py implemented (NW alignment + SME scoring)
 - ✓ Smoke test passed (5 incidents end-to-end through A2 pipeline)
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Next Steps

- 1 Build full v2 cache (~2,243 incidents)
- 2 Run A0 vs A2 on full test set — compare metrics
- 3 Tune alpha on training data

References

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